SUMS OF DIVISORS ON A CURVE ARE FINITE LOCALLY FREE OF ADDITIVE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 677-683 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 685-723, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_pic.json (pb=1641, stmt 677-683, proof 685-723).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 02MC: sources/stacks-pic/pinned-original-source.tex, source lines 723–723. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B8U: sources/stacks-pic/pinned-original-source.tex, source lines 723–723. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9D: sources/stacks-pic/pinned-original-source.tex, source lines 625–634. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LS: sources/stacks-pic/pinned-original-source.tex, source lines 723–723. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02KB: sources/stacks-pic/pinned-original-source.tex, source lines 723–723. Cited by the retained statement or proof but not part of the

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collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03GN: sources/stacks-pic/pinned-original-source.tex, source lines 723–723. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09MQ: sources/stacks-pic/pinned-original-source.tex, source lines 723–723. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$ be a smooth morphism of schemes of relative dimension 1. Let $D_1, D_2 \subset X$ be closed subschemes finite locally free of degrees d_1, d_2 over S. Then $D_1 + D_2$ is finite locally free of degree $d_1 + d_2$ over S.*

Proof. By Lemma 0B9D we see that D_1 and D_2 are relative effective Cartier divisors on X/S. Thus $D = D_1 + D_2$ is a relative effective Cartier divisor on X/S by Divisors, Lemma 0B8U. Hence $D \to S$ is locally quasi-finite, flat, and locally of finite presentation by Lemma 0B9D. Applying Morphisms, Lemma 09MQ the surjective integral morphism $D_1 \amalg D_2 \to D$ we find that $D \to S$ is separated. Then Morphisms, Lemma 03GN implies that $D \to S$ is proper. This implies that $D \to S$ is finite (More on Morphisms, Lemma 02LS) and in turn we see that $D \to S$ is finite locally free (Morphisms, Lemma 02KB). Thus it suffice to show that the degree of $D \to S$ is $d_1 + d_2$. To do this we may base change to a fibre of $X \to S$, hence we may assume that $S = \mathrm{Spec}(k)$ for some field k. In this case, there exists a finite set of closed points $x_1, \ldots, x_n \in X$ such that D_1 and D_2 are supported on $\{x_1, \ldots, x_n\}$. In fact, there are nonzerodivisors $f_{i,j} \in \mathcal{O}_{X,x_i}$ such that

$$D_1 = \coprod \mathrm{Spec}(\mathcal{O}_{X,x_i}/(f_{i,1})) \quad \text{and} \quad D_2 = \coprod \mathrm{Spec}(\mathcal{O}_{X,x_i}/(f_{i,2}))$$

Then we see that

$$D = \coprod \mathrm{Spec}(\mathcal{O}_{X,x_i}/(f_{i,1}f_{i,2}))$$

From this one sees easily that D has degree $d_1 + d_2$ over k (if need be, use Algebra, Lemma 02MC). □